

When discussing regions in 2-space, we observed that there were multiple ways to ‘view’ the region. Specifically, we could use the Cartesian coordinate system *or* the polar one. However, we’ve also seen that triple integrals over closed and bounded regions in 3-space are also useful. Thus, we ask “Are there multiple ways to ‘view’ regions in 3-space?” The answer is yes! In this discussion we introduce two additional ways.

We start with a system that is a vertical extension of polar coordinates:

Definition. **Cylindrical coordinates** represent a point P by an ordered triple (r, θ, z) , where $r \geq 0$,

- (i) (r, θ) is a polar representation of the projection of P in the xy -plane, and
- (ii) z is the directed, vertical distance from (r, θ) to P .

Since this is literally just the vertical extension of polar coordinates, we have (x, y, z) and (r, θ, z) related by the following equations:

Cylindrical To Rectangular:

$$x = r \cos(\theta), \quad y = r \sin(\theta), \quad z = z, \quad r^2 = x^2 + y^2, \quad \text{and} \quad \tan(\theta) = \frac{y}{x}$$

Theorem. [Fubini’s Theorem (Cylindrical Form)] Let $F(x, y, z)$ be a continuous function on the closed and bounded region D with cylindrical representation $F(r, \theta, z) = F(r \cos(\theta), r \sin(\theta), z)$. Using a ‘ $dz \, dr \, d\theta$ ’ ordering,

$$\iiint_D F(x, y, z) \, dV = \int_{\theta=\alpha}^{\theta=\beta} \int_{r=h_1(\theta)}^{r=h_2(\theta)} \int_{z=g_1(r,\theta)}^{z=g_2(r,\theta)} F(r, \theta, z) r \, dz \, dr \, d\theta.$$

Example. Consider the closed and bounded region D bounded below by the plane $z = 0$, laterally by the cylinder $x^2 + (y - 1)^2 = 1$, and above by the paraboloid $z = x^2 + y^2$. Setup an integral that returns the volume of D .

Example. Consider the closed and bounded region D bounded below by the cone $z = \sqrt{x^2 + y^2}$ and above by the paraboloid $z = 2 - x^2 - y^2$. Setup an integral that returns the volume of D .

The next coordinate system we will encounter is a ‘targeting’ coordinate system:

Definition. **Spherical coordinates** represent a point Q by an ordered triple (ρ, ϕ, θ) ,¹ where

- (i) ρ is the distance between Q and the origin (with $\rho \geq 0$),
- (ii) θ is the angle used in cylindrical coordinates for $r \geq 0$, and
- (iii) ϕ is the angle between the positive z -axis and the line segment $\overline{OQ}$ (with $0 \leq \phi \leq \pi$).

Spherical To Cylindrical:

$$r = \rho \sin(\phi), \quad z = \rho \cos(\phi), \quad \theta = \theta, \quad \text{and} \quad \rho = \sqrt{r^2 + z^2}$$

Spherical To Rectangular:

$$x = \rho \sin(\phi) \cos(\theta), \quad y = \rho \sin(\phi) \sin(\theta), \quad z = \rho \cos(\phi), \quad \text{and} \quad \rho = \sqrt{x^2 + y^2 + z^2}$$

Example. Find a spherical coordinate equation for the sphere $x^2 + y^2 + (z - 1)^2 = 1$.

¹Note that physicists interchange the roles of θ and ϕ when doing calculations in spherical coordinates.

Example. Find a spherical coordinate equation for the cone $z = \sqrt{x^2 + y^2}$

Theorem. [Fubini's Theorem (Spherical Form)] Let $F(x, y, z)$ be a continuous function on the closed and bounded region D with spherical representation

$$F(\rho, \phi, \theta) = F(\rho \sin(\phi) \cos(\theta), \rho \sin(\phi) \sin(\theta), \rho \cos(\phi)).$$

Using a ' $d\rho d\phi d\theta$ ' ordering,

$$\iiint_D F(x, y, z) dV = \int_{\theta=\alpha}^{\theta=\beta} \int_{\phi=h_1(\theta)}^{\phi=h_2(\theta)} \int_{\rho=g_1(\phi,\theta)}^{\rho=g_2(\phi,\theta)} F(\rho, \phi, \theta) \rho^2 \sin(\phi) d\rho d\phi d\theta.$$

Example. Find the volume of the 'ice cream cone' D cut from the solid sphere $\rho \leq 1$ by the cone $\phi = \frac{\pi}{3}$.

Example. Consider the region D bounded below by the hemisphere $\rho = 1$, $z \geq 0$, and above by the cardioid of revolution $\rho = 1 + \cos(\phi)$. Find the volume of D .

Example. Consider the region D bounded below by the xy -plane, on the sides by the sphere $\rho = 3$, and above by the cone $\phi = \frac{\pi}{3}$. Find the volume of D .